

Part2: What is the difference between the following objects in SQL Server

■ batch, script and transaction

Batch	Script	Transaction
A Set of queries executed together but no one of them affects the result of any other query. Some of them may fail and doesn't affect others	A Set of queries only can be executed together when separated with "GO" keyword	A Set of queries must be executed either all together or none of them executed at all. All as one unit, if one failed The rest transaction rolled back then,

■ trigger and stored procedure

Stored Procedure	Trigger
A piece of code by developer to present a specific design, executed when called.	a special type of stored procedure structure, executed explicitly when a related event occurs.

■ stored procedure and functions

Stored Procedure	Function
-has an only one and integer data typed return value -receive input, output and input-output parameters. -Contains DML queries -only called in select phrase in a query.	-has return value(s) which can be represented by any data type. -receive only input parameter -Contains only Select statements. -can be called in any phrase in the query.

■ drop, truncate and delete statement

Drop	Truncate	delete
-DDL query -Remove the table from the DB (delete table's data and metadata)	-DDL query -Remove the table 's data but keeps table's metadata. - Does reset table's identity column values.	-DML query -Remove a certain data from the table by where condition. - Doesn't reset table's identity column values.

■ select and select into statement

Select	Select into
To retrieve data from table.	To insert selected data into a new table

■ local and global variables

Local Variable	Global Variable
-Starts with @ -can be declared and assigned as per developer -has local scope (batch, function, sp)	-Starts with @@ -Can't be declared nor assigned -has global scope and affected by last query

■ convert and cast statements

Convert	Cast
Used for converting a variable data type from one to another, accepts optional formatting argument	Used for converting a variable data type from one to another.

1. DDL,DML,DCL,DQL and TCL

DDL	DML	DCL	DQL	TCL
Data Definition Language	Data Manipulation Language	Data Control Language	Data Query Language	Transaction Control Language
SQL command used to define descriptions of the database schema and to create and modify the structure of database objects in the database. (create, drop, alter...etc)	SQL command used to deal with the data manipulation in the database and to control the data access and modifying (insert, update, delete...etc)	SQL commands used to deal with the permissions and other controls of the database system. (grant, deny and revoke)	SQL command used to make queries on the data (Select)	SQL command used to control the transactions in database (commit, Rollback .. etc)

■ For xml raw and for xml auto

XML Raw	XML Auto
Used in process to represent data from SQL to xml File By default, represent columns of table as tag attributes, “ELEMENTS” clause can be used to represent them in separated tags	Used in process to represent data from SQL to xml File Used to represent joined tables

■ Table valued and multi statement function

Table Valued	Multi Statement
Generally, returns a table as a return value which is structured based on selected statement values	Returned table generally specified by return syntax

■ Varchar (50) and varchar(max)

Varchar(50)	Varchar(max)
Character data type. Used with character data type with maximum of 50 Used when the input size is almost specified	Column will store the entered value and resize itself upon it. Used when the input size couldn't be specified.

Datetime(3), datetime2(7) and datetimeoffset(7)

Datetime(3)	Datetime2(7)	Datetimeoffset(7)
- Size: 8 bytes - Char length: 19 - Date range is 1753-01-01 through 9999-12-31. - Time range is 00:00:00 through 23:59:59.997. - No Time Zone Offset.	- Size :6 to 8 bytes, depends on precision + 1-byte stores precision. - Char length: 19 - Date range is 0001-01-01 through 9999-12-31. - Time range is 00:00:00 through 23:59:59.9999999 - No Time Zone Offset	-Size :8 to 10 bytes, depends on precision + 1-byte stores precision - Char length: 26 - Date range is 0001-01-01 through 9999-12-31 - Time range is 00:00:00 through 23:59:59.9999999 - Time zone offset

■ Default instance and named instance

Default Instance	Named Instance
- Installed by default when installing a single instance of SQL server. - Only one default instance	- When Naming the Instance. - Could be multiple named instances

■ SQL and windows Authentication

Windows Auth.	SQL Auth
Authentication of the administrator of the Operating system, so SQL Server on this operation system doesn't require any additional confirmation to give access	There could be many sql users on the same server differ from the windows authorized users, so it is set to require password, in addition it can be connected remotely from the server.

■ Clustered and non-clustered index

Clustered Index	Non Clustered Index
- Created by default on primary key column so it's on the actual data. -Faster than non-clustered index - Better towards memory wise -only one clustered index - In Clustered index leaf nodes are actual data itself. - In Clustered index, Clustered key defines order of data within table. - A Clustered index is a type of index in which table records are physically reordered to match the index.	-Table can have multiple non-clustered indexes. -Slower than clustered index. -stores pointers to actual data - In Non-Clustered index, index key defines order of data within index. - A Non-Clustered index is a special type of index in which logical order of index does not match physical stored order of the rows on disk.

■ Group by rollup and group by cube

Group by rollup	Group by cube
used to calculate sub-totals and grand totals for a set of columns passed to the "GROUP BY ROLLUP" clause. Can also be used to calculate sub-totals for each column, based on grouping.	CUBE operator produces results by generating all combinations of columns specified in the GROUP BY CUBE clause.

■ Inline function and view

Inline	View
Have return value, Takes Parameters	No parameters Automate many complex, repeatable queries Can be used by server engine optimizer

■ Table variable and temporary table

Table variable	Temp Table
Has Variable scope and life time	Physical table created in tempdb
Not physical but memory located table	Lifetime depends on its type
Can be passed to function or stored procedure	Can have indexes and constraints
	Can't passed to function or sp

■ Row_number() and dense_Rank() function

Row_number()	Dense_rank()
Creates unique rank however replicated records	Creates unique ranks but replicated records